

MANIKANDAN S

AI/ML Engineer

CS student with hands-on experience in AI/ML, LLM Engineering, and backend development.

Focused on building intelligent systems that are both efficient and interpretable.

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EDUCATION

SASTRA Deemed University | B.Tech, Computer Science (IoT & Automation) 2023 - 2027

Trichy, Tamil Nadu · CGPA: 7.46 / 10.0

Relevant Coursework: Machine Learning, Deep Learning, Data Structures, Operating Systems, Computer Networks, DBMS

TECHNICAL SKILLS

Languages: Python, C, C++, Java

Systems: Linux, Bash, Docker, Git/GitHub, AWS (EC2, S3), CI/CD (GitHub Actions), Networking Basics

AI / ML: NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, ML Pipelines, Evaluation Metrics, Transformers

LLM Engineering: OpenAI / Gemini / Claude APIs, Prompt Engineering, Function Calling, Embeddings, Vector Search (FAISS, Chroma), RAG

AI Frameworks: LangChain, LangGraph, Streamlit, Gradio

Backend: FastAPI, REST APIs, Flask, SQL, JSON, Postman

RESEARCH PUBLICATIONS

Knowledge Distillation for Efficient and Lightweight Cancer Classification

Manikandan S, Aniruddh S B, Balaji B P · IEEE Journal · [Under Review](#)

Distilled a soft-voting Teacher ensemble (ResNet50, InceptionV3, DenseNet121) into a lightweight MobileNetV2 Student using KL-divergence loss and Offline Histogram Matching — achieving 97.90% accuracy on 25,000 histopathological images with ~80% model size reduction (76.9MB → 3.5MB).

PROJECTS

Faith-XAI | Python, TensorFlow, Flask, SHAP

github.com/mani30mk/Faith-XAI

Explainable AI system for loan risk assessment trained on the Home Credit Default Risk dataset (307,511 records), deployed via Flask REST API.

- Designed and trained a deep learning binary classifier to predict loan default risk with probability scores.
- Integrated SHAP-based explainability to surface the top-5 feature drivers influencing each prediction.
- Engineered custom trust metrics — Faithfulness (deletion AUC) and Robustness (sensitivity analysis) — to quantify explanation reliability beyond standard accuracy.
- Generated counterfactual suggestions for high-risk applicants, showing minimal input changes to flip the prediction outcome.
- Served the full pipeline via a Flask REST API with an interactive dark-themed web dashboard and XEQ user feedback system.

Hybrid Self-Evolving Neural Network | *Python, TensorFlow, Flask, AutoML*

github.com/mani30mk/Hybrid-Self-Evolving-Neural-Network

AutoML system that generates, optimizes, and trains deep learning architectures from scratch using only a user prompt and dataset, served via Flask REST API.

- Designed an evolutionary search pipeline that auto-grows network depth and selects optimal layer types through mutation and crossover mechanisms.
- Automated input shape correction and architecture validation, eliminating manual model design across image classification, sequence, and regression tasks.
- Exposed architecture generation and training workflows through RESTful API endpoints for programmatic access.

EXTRACURRICULAR ACTIVITIES

- Manages a self-hosted Linux home server for AI/ML project deployment; prototypes with 3D printing and CAD tools.
- Engages with academic research in AI, physics, and computational mathematics.

COURSES

Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp

Udemy

LANGUAGES

Tamil (Native) · **English** (Full Professional Proficiency)